

# Yahya Mirza

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## EDUCATION

### University of California, San Diego (Junior)

Bachelor of Science in Mechanical Engineering

**Coursework:** Fluid Mechanics, Thermodynamics, Solid Mechanics, Statics, Dynamics, Linear Circuits, Machining, MATLAB

La Jolla, CA

Sep. 2025 – June 2027

### Foothill College

Mechanical Engineering for Transfer (GPA: 4.0)

**Coursework:** SOLIDWORKS, Linear Algebra, Differential Equations, Vector Calculus

Los Altos Hills, CA

Sep. 2024 – Aug 2025

## SKILLS

Mechanical Design · SOLIDWORKS · GD&T · OnShape · Fusion 360 · Arduino · ESP32 · Raspberry Pi · C++ · MATLAB · FEA Analysis · Machining · Git · VS Code · CNC Milling · Risk Reduction · Technical Writing · AutoCAD · DFM/DFA · Soldering

## EXPERIENCE

### Robotics Engineer – Yonder Dynamics (Competition Mars Rover Team)

Sep. 2025 – Present

University of California, San Diego

La Jolla, CA

- Led an **end-to-end** production of a **linear slider mechanism** capable of reliably translating a 30lb, 5-axis robotic arm with a **range of motion of 10"**.
- Developed a comprehensive and **parametric CAD model** of the module in **OnShape**, implementing **DFMEA** procedures to ensure **cheap**, **realistic**, and **quick** fabrication.
- Verified via **static analysis calculations** (with **5x FOS**) that the module powered with a NEO550 brushless motor coupled with a 50:1 Harmonic Drive can reliably translate a **200N payload** at an angle of 60 degrees from the horizontal.
- Fabricated and assembled over **20+ parts** via **milling (manual and CNC)**, **water jetting**, **3D printing**, **laser cutting**, and **sand blasting**, ensuring **safe** and reliable machining practices.
- Planned **validation activities** such as determining **max load**, suitable **operation temperatures**, and **endurance** capabilities, cooperating with necessary subteams for easy system integration.
- Documented the project in a dedicated **engineering report** containing detailed diagrams and **FBD's**, calculations, motivations, iterations, etc.

### Mechanical Design Intern – Nanotech @ Berkeley Laboratory

Jun. 2025 – Oct 2025

University of California, Berkeley

Berkeley, California

- Prototyped and fabricated (3D Printing) a **stepper motor** driven assist for lateral shoulder movement in **SOLIDWORKS**, mitigating fluctuations in EEG neural stimulation by **20%**
- Developed and integrated circuitry for 4 **Nema 17** stepper motors onto an **Arduino Nano** utilizing **C++** that allow for **independent** motor control.
- Designed, modified, and fabricated **15+ parts** for arm, shoulder, and ankle modules for an EEG controlled exoskeleton, significantly accelerating the development of the project.

### Project Co-lead – FireFlight RC Plane

Sep. 2024 – Sep. 2025

Foothill College Engineering Club

Los Altos Hills, CA

- Designed and refined the **tail assembly** and **elevon** control surfaces of Foothill College's **FireFlight RC Plane** in **SolidWorks** and **OnShape**; optimized aerodynamics through iterative prototyping, 3D printing, and testing, decreasing the **drag coefficient** by over **17%** and increasing **lift** by **30%**
- Directed a **10+ member** team across airframe, electronics, and control systems; organized weekly meetings and increased member participation by **3x**.
- Delivered **technical presentations** at the **2025 Bay Area Honors Research Symposium** (UC Berkeley) and Foothill College's **RSLS Symposium**, where the team earned **first place** among **162 groups**.

## PROJECTS

### Competition Soccer Robot | Fusion 360, Statics, Machining, Circuitry

Oct 2025 – Dec 2025

- Successfully built a competition robot designed to sweep and score 4" dia. soccer balls into opponent's goal. Developed a **two-stage, belt-driven flywheel kicker mechanism (9:1 reduction)** resulting in an average **kicking distance** of **70%** of the span of the field, **rapidly iterating** via risk reduction procedures.
- Developed a comprehensive CAD model in **Fusion 360**, notably **designing** the robot's **swiping arm subassembly**
- Validated** the arm module with a **6x FOS** via static analysis and **experimental verification**, authoring a **technical paper** documenting results.
- Authored a **technical presentation** demonstrating the functions of the robot and detailing **risk reduction** and analysis procedures, receiving a nomination for **best technical presentation** out of **34 teams**.
- Awarded** for **"Best Manufacturing"**; chosen by top engineering professors and competition judges.